

Radiation: $\Rightarrow$ Heat transfer via electromagnetic waves
No need for a medium $\Rightarrow$ not diffusion

Examples:

- 1) Sun heating the earth
- 2) Radiative heaters in store entrances & theaters
- 3) IR (infrared) cameras & detection

Fourier's Law of Heat Conduction

In 17th century, a guy named Joseph Fourier discovered an empirical correlation that governs heat transfer in a solid:

$$\boxed{q_x'' = -k \frac{dT}{dx}}$$

$\uparrow$ Heat flux in the x direction $\uparrow$ Thermal Conductivity of the material transferring heat Δ Temperature gradient in the x-direction. Ordinary derivative

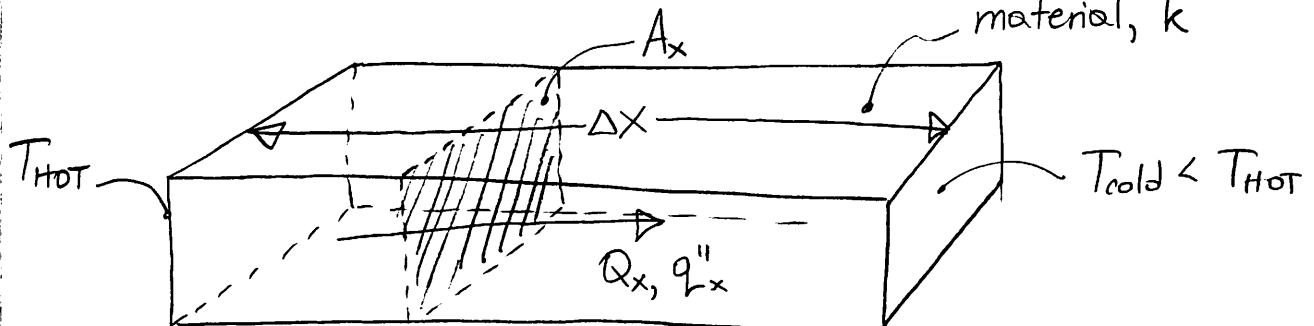
Sometimes written as:

$$\boxed{Q_x = -k A_x \frac{dT}{dx}}$$

$\uparrow$ Cross sectional area perpendicular to heat transfer dir.

$$\frac{dT}{dx} = \frac{T_{\text{cold}} - T_{\text{hot}}}{x}$$

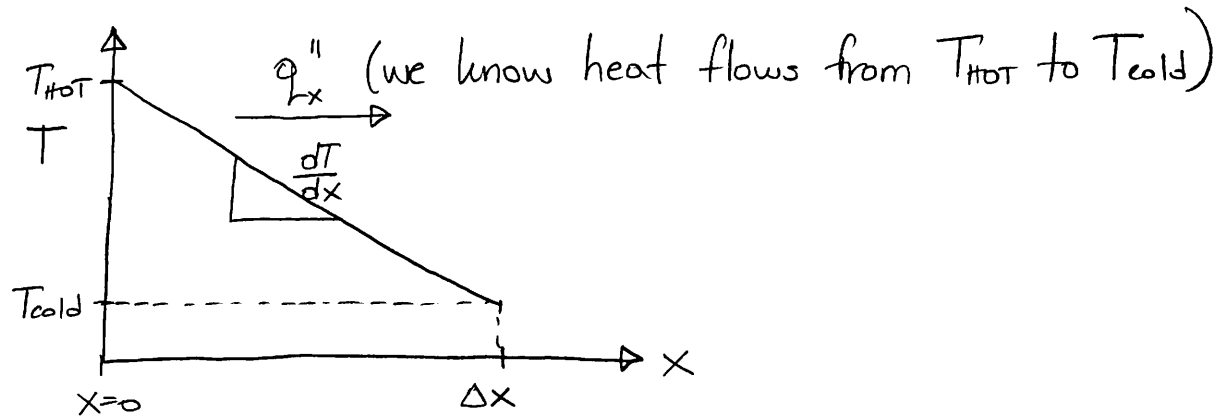
material, k



Fourier did this experiment for many metals to obtain his relation. (4)

Let's take a closer look at his equation:

$$q''_{Tx} = -k \frac{dT}{dx}$$

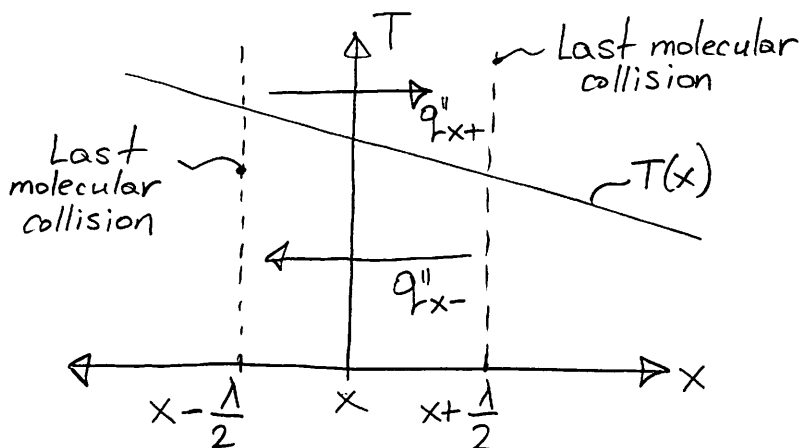


$$\frac{dT}{dx} = \frac{T_{cold} - T_{hot}}{\Delta x} < 0 \text{ (negative)}$$

But q''_{Tx} must be positive since heat flows in the positive x -direction?

↳ Hence the negative sign in the equation.
 k is always a positive quantity, hence things make sense. Heat flows down the temperature gradient.

Derivation of Fourier's Law



$n \equiv \# \text{ of particles/m}^3$
 $m \equiv \text{mass per particle}$
 $C_p \equiv \text{specific heat capacity}$
 $\bar{c} \equiv \text{average speed of particle}$

$C_p [\text{J/kg} \cdot \text{K}]; n [\#/\text{m}^3], m [\frac{\text{kg}}{\#}]$

Lets look at the energy exchange at our plane of interest (x)

$$q''_{L_{x+}} = n \cdot m \cdot c_p \cdot \bar{c} \cdot T\left(x - \frac{\lambda}{2}\right) \quad \text{Temperature at } x = x - \frac{\lambda}{2}$$

$$\left[\frac{\#}{m^3}\right] \cdot \left[\frac{kg}{\#}\right] \cdot \left[\frac{J}{kg \cdot K}\right] \cdot \left[\frac{m}{s}\right] \cdot [K]$$

$$\underbrace{\left[\frac{kg}{m^2 \cdot s}\right]}_{\text{mass flow rate of particles per unit area.}} \quad \underbrace{\left[\frac{J}{kg}\right]}_{\text{energy flow per unit mass of particles.}}$$

We can notice that: $n \cdot m = \rho \left[\frac{kg}{m^3}\right] \Rightarrow \text{Density}$

$$q''_{L_{x+}} = + \rho c_p \bar{c} T\left(x - \frac{\lambda}{2}\right)$$

$$q''_{L_{x-}} = + \rho c_p \bar{c} T\left(x + \frac{\lambda}{2}\right)$$

$$q''_{L_x} = q''_{L_{x+}} - q''_{L_{x-}} = \rho c_p \bar{c} \left[T\left(x - \frac{\lambda}{2}\right) - T\left(x + \frac{\lambda}{2}\right) \right]$$

$$= - \rho c_p \bar{c} \left[T\left(x + \frac{\lambda}{2}\right) - T\left(x - \frac{\lambda}{2}\right) \right]$$

Multiply the right hand side by (λ/λ) :

$$q''_{L_x} = - \rho c_p \bar{c} \lambda \left[\frac{T\left(x + \frac{\lambda}{2}\right) - T\left(x - \frac{\lambda}{2}\right)}{\lambda} \right]$$

$$\underbrace{\hspace{10em}}_{\frac{dT}{dx}}$$

$$\boxed{q''_{L_x} = - \rho c_p \bar{c} \lambda \frac{dT}{dx}}$$

$$k = \rho c_p \bar{c} \lambda \Rightarrow \left[\frac{kg}{m^3}\right] \cdot \left[\frac{J}{kg \cdot K}\right] \left[\frac{m}{s}\right] [m] = \frac{J}{m \cdot s \cdot K} = \frac{W}{m \cdot K}$$

Typical Values

$$k_{air} \approx 0.1 \text{ W/m} \cdot K$$

$$k_{water} = 0.6$$

$$k_{polymer} \approx 0.1$$

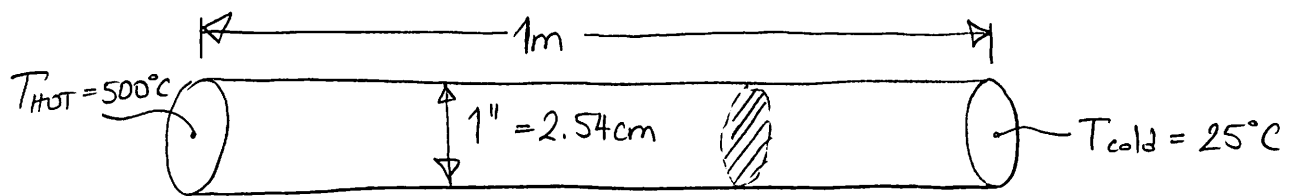
$$k_{Cu} = 400$$

$$k_{Al} = 300$$

W

Example | Conduction through a bar:

A bar of copper ($k_{cu} = 400 \text{ W/m} \cdot \text{K}$) contacts a hot source at 500°C on one end. The opposite end of the bar is cooled to ambient ($T_{\text{cold}} = 25^\circ\text{C}$) by cooling water. The length of the bar is 1m , and it is cylindrical with a diameter of $1''$. What is the heat transfer rate? What is the heat flux?



$$q'' = -k \frac{dT}{dx} = -(400 \text{ W/m} \cdot \text{K}) \cdot \left(\frac{25^\circ\text{C} - 500^\circ\text{C}}{1\text{m}} \right)$$

$$= 190\,000 \text{ W/m}^2$$

$$\boxed{q'' = 19 \text{ W/cm}^2}$$

$$Q = -kA \frac{dT}{dx} = q''A = (19 \text{ W/cm}^2) \left(\pi \frac{(2.54\text{cm})^2}{4} \right)$$

$$\boxed{Q = 96.2 \text{ W}}$$

So how large is this?

$$q''_{\text{boiling}} \approx 100 \text{ W/cm}^2$$

$$q''_{\text{nat. convection}} \approx 0.01 \text{ W/cm}^2$$

Think about power plants: $\dot{W} = 1000 \text{ MW}$ electrical output
Assuming efficiency of 50% ($\eta = 0.5$)

$$\eta = \frac{\dot{W}}{Q_{\text{in}}} \Rightarrow \boxed{Q_{\text{in}} = 2000 \text{ MW} = 2\,000\,000\,000 \text{ W!}}$$

The previous example was a good demonstration of Fourier's law, but it cannot be used for more complex situations:

- Transient heat transfer
- Heat loss to the air along the length of the bar
- Radiation effects
- Heat generation.

We need a more general formulism to handle these problems.

Conduction Heat Transfer (Chapter 2 of Textbook)

The equation of conduction heat transfer (heat diffusion eqn) is the first law of thermodynamics subject to the following assumptions:

- 1) No mass flow in or out of the control volume (CV) or system.
- 2) No change in Kinetic or Potential energy
- 3) There may exist a heat generation term, eg. I^2R from electrical losses, or radioactive decay, etc...

